

# Quantitative Trading with Python: A Top-Down Approach

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# Chapter 1

## Codes of Life

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### 1.1 Data vs. Information

The advent of digital technologies at the turn of the 21st century fundamentally augmented humanity’s relationship with the world around us. This revolution kickstarted the modern understanding of “data” as a fundamental resource for decision-making, communication, and innovation. What began as simple digitization—the conversion of analog information into binary form—quickly evolved into something far more expansive. As sensors proliferated, connectivity spread, and storage costs plummeted, we crossed a threshold: the era of *big data*.

The term “big data” describes not just the sheer volume of information being generated, but the unprecedented velocity at which it arrives and the variety of sources from which it flows. Social media posts, smartphone location traces, financial transactions, scientific instruments, surveillance cameras—each generates continuous streams of data points that accumulate at scales previously unimaginable. The ideas surrounding big data and its potential for transforming our wider society are now pervasive in modern discourse. The thoughts and grand ambitions out of Silicon Valley continue to make the case for data as the new frontier of economic and social transformation—the proverbial gold rush of the digital age.

In response to this data-driven shift, entire fields have emerged to help us understand and interpret the data we encounter and choose to collect. Disciplines such as statistics, computational thinking, data science, and machine learning offer tools and frameworks for extracting meaning from vast datasets. Yet to work with data effectively, we must first understand its fundamental nature. Consider how you read and interpret the very text before you: *How do you interpret the symbols on this page?*

An easy answer might be to say “the alphabet,” but consider what it means to *understand* the alphabet in the context of this text—or really, any text. Consider the following sequence of symbols:

## THEOSTRICHCANNOTFLY

You likely parsed that instantly: *The ostrich cannot fly*. But nothing in the string itself told you where one word ends and another begins—no spaces, no punctuation, no explicit boundaries of any kind. You supplied the rules yourself: word boundaries, grammatical structure, semantic plausibility, drawing on a lifetime of learned context applied so automatically you may not have noticed doing it.

Now consider the same symbols, differently arranged:

## THEO STRICH CAN NOT FLY

Suddenly: *Theo Strich cannot fly*. The data is identical. A single structural rule—where spaces appear—has changed the meaning entirely. The production of data is easy; the challenge lies in interpreting and deriving **meaning** from it.

Consider the example of three civilizations, each with its own varying complexities in how data was transcribed.

The first would be humanity’s humble beginnings, where early humans used simple symbols and drawings to communicate and record their experiences. This is largely considered a key marker of when humans began to think in abstract terms—a precursor to the development of complex language and writing systems.

These early forms of data were carefully drawn symbols etched into caves, walls, or stone. A 2026 finding on the Indonesian island of Sulawesi dates back nearly 67,800 years. These handprints and animal silhouettes begat questions humanity fundamentally will likely never answer: *why*. We see it and assume it was a signature, a marker of presence—“I was here.” But was it? Was it a ritual? A warning? A record of a hunt? A spiritual practice?

This is the fundamental fragility of data: without context, it becomes a guessing game. The cave painter had perfect information—they knew exactly what they meant, if they meant anything at all. But we, the interpreters separated by millennia, possess only the data. We can analyze pigment composition, dating techniques, spatial arrangement, yet perhaps the most important question of all remains trapped in silence.

Such systems naturally evolve with the needs and capabilities of the societies that create them. We see this evolution in the transition from cave paintings to hieroglyphs invented by the ancient Egyptians.

Unlike the ambiguous handprints of Sulawesi, hieroglyphs were *designed*. They arose from a deliberate need: to record and communicate administrative and religious information with standardization and clarity. Here was humanity’s first attempt to solve the context problem—to create a system where the same symbol would mean the same thing to everyone who encountered it.

This was done through *organization* and *structure*. Consider two identical sets of hieroglyphs

arranged differently as shown in Table 1.1.

Table 1.1: Identical hieroglyphs gain different meanings through structural context

| Context/Structure         | Same Glyph                                                                                          | Resulting Meaning                |
|---------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------|
| Isolated on temple wall   |  (Eye of Horus)    | Ritual symbol, divine protection |
| In chronological sequence |  (position 3 of 5) | “Third event: the watching”      |
| In tax ledger column      |  (quantity: 12)    | “12 units of temple offering”    |

An individual hieroglyph carved in isolation tells us almost nothing. But a collection of hieroglyphs arranged in sequence, organized by rules, structured in rows and columns—that structure creates meaning. The data is identical. The structure transforms it into different information depending on how it’s collected and organized. What we choose to measure, when we measure it, and under what conditions—along with how we organize and encode relationships within the data—fundamentally determine its meaning and usefulness.

The key understanding is that data is not inherently meaningful; it requires context, organization, and structure to become **information**.

#### Definition — Data vs. Information

- **Data:** Raw symbols, measurements, or observations without inherent meaning
- **Information:** Data + structure + context + interpretation

This is the critical insight: **information—not raw data—is what ultimately drives decisions**.

Raw data sitting in a warehouse, uncollected with care or organized without intention, is inert—incapable of informing decisions on its own.

Information—data that has been thoughtfully collected, properly structured, and interpreted within context—is what actually moves decisions, drives strategy, and creates value.

The difference between raw data and actionable information is often the difference between failure and success. This textbook will impart the principles and practices of data management, collection, analysis, visualization, and interpretation. In doing so, it will also challenge readers to ask fundamental questions about the role of data in our lives and the world around us. Without this understanding, we cannot fully appreciate the power and potential of data in shaping our world. Without it, we risk causing incredible harm by misinterpreting, incorrectly collecting, or misusing data.

Each chapter will explore real-world applications of these concepts through the three civilizations we have examined. The final civilization—the one navigating the data deluge of the 21st century—is *our own*.

# Bibliography